

CLASS 02 / SEP 1 / U02

TRANSFORMATIONS AND CSG

P1 BEGINS — DESCRIBE A SOLID AS AN EDITABLE SEQUENCE

NOTHING DUE

P1a ISSUED TODAY

WHY NON-DESTRUCTIVE

The same two solids, subtracted. One keeps its history; one throws it away.

DESTRUCTIVE

CUBE - SPHERE → ONE MESH

The sphere is gone. Its radius is gone. To change anything you rebuild from scratch.

NON-DESTRUCTIVE

CUBE + SPHERE → SUBTRACT → RESULT

Every input is still a live node. Change the radius and everything downstream recomputes itself.

P1a REQUIRES YOUR OPERATION HISTORY TO BE INTACT. YOU WILL BE ASKED TO CHANGE A PARAMETER IN FRONT OF THE CLASS.

TRANSFORMATIONS ARE ARITHMETIC

Three operations you already know, written as what they actually do.

MOVE

= ADD

$$x' = x + dx$$

$$y' = y + dy$$

$$z' = z + dz$$

$$(2, 1) + (3, 0) \rightarrow (5, 1)$$

SCALE

= MULTIPLY

$$x' = x \times f$$

$$y' = y \times f$$

$$z' = z \times f$$

$$(2, 1) \times 2 \rightarrow (4, 2)$$

ROTATE

= MIX x AND y

$$x' = x \cdot \cos\theta - y \cdot \sin\theta$$

$$y' = x \cdot \sin\theta + y \cdot \cos\theta$$

$$(1, 0) \text{ rot } 90^\circ \rightarrow (0, 1)$$

Move and scale treat each coordinate on its own. Rotation is the only one where x and y have to talk to each other.

THREE BOOLEANS

Everything in P1a is built from these, applied in order.

UNION

$$A + B$$

Everything either solid occupies.

DIFFERENCE

$$A - B$$

A, with B carved out of it.

INTERSECTION

$$A \cap B$$

Only where both solids overlap.

ORDER MATTERS. $A - B$ IS NOT $B - A$, AND YOUR DIAGRAM MUST MAKE THE OPERANDS UNAMBIGUOUS.

Reversing a subtraction is one of the three comparisons P1a asks for. Do it once deliberately and you will never confuse the operands again.

THE SEQUENCE

What we build together, in order, this session.

01

Introduce the node editor, inputs, outputs and parameters using one primitive.

02

Move, rotate and scale procedurally; change the order and compare the result.

03

Compare union, difference and intersection between two solids, including reversing the subtraction.

04

Expose useful inputs and compose a short massing sequence.

THE NOTATION

Write the procedure as pseudocode. Formal enough to be unambiguous, plain enough to read aloud.

THE OPERATIONS

```
box(x, y, z)      cylinder(r, h)
sphere(r)         cone(r, h)

move(s, x, y, z)  rotate(s, axis, deg)
scale(s, factor)

union(a, b)       subtract(a, b)
intersect(a, b)
```

WORKED EXAMPLE

```
base  = box(6, 4, 3)
tower = box(2, 2, 6)
tower = move(tower, 2, 0, 3)
void  = cylinder(r=1, h=5)
void  = move(void, -1.5, 0, 0)

mass  = union(base, tower)
result = subtract(mass, void)
```

THREE RULES: EVERY SHAPE IS BORN AT THE ORIGIN · UNITS ARE ABSTRACT · EVERY STEP ASSIGNS A NAME, SO A SUBTRACTION IS NEVER IN DOUBT.

THE SAME THING, WRITTEN OUT

You are not asked to type this. Read it, and check that it says what the nodes you just built say.

```
base  = box(6, 4, 3)
tower = box(2, 2, 6)
tower = move(tower, 2, 0, 3)

void  = cylinder(r=1, h=5)
void  = move(void, -1.5, 0, 0)

mass   = union(base, tower)
result = difference(mass, void)    # not difference(void, mass)
```

WHAT IT MAPS ONTO

parameter
vector
translation
rotation
scale
union
difference
intersection

THIS IS THE NOTATION FROM THE LAST SLIDE, AND IT IS ALSO PYTHON. MORE ON THAT IN DECEMBER.

ISSUED TODAY

P1a — CSG MASSING SEQUENCE

DUE WEEK 4 — Sep 15 / 7%

Build a parametric massing study, reconstruct its cutter from basic geometry, and develop a small printable variant. This first milestone is the massing study: an ordered sequence of primitives and Booleans, diagrammed so another person could follow it.

STAGE 1

THE SEQUENCE

Primitives, transformed and combined in a stated order.

STAGE 2

THE DIAGRAM

One drawing per operation, plus a final axonometric.

STAGE 3

THE VARIANTS

Three parameter settings off the same graph.

REQUIREMENTS — P1a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

One 17 × 11 inch sheet with the ordered transformation/Boolean diagram, a final axonometric and three parameter variants.

02

Demonstrate translation, rotation and scale, and compare union, difference and intersection. Clearly identify the operands of subtraction.

03

Submit an editable graph with exposed inputs and geometry generated in the graph. Keep the source operations intact.

NOW — BUILD THE SEQUENCE

Rest of the session: a working graph with something you can change in front of me.

✓

Object chosen — orthogonal massing, 5–10 clear moves, no complex curves

✓

First three primitives placed, each born at the origin and moved from there

✓

At least one Boolean working, with the operands named

✓

Two useful inputs exposed, so a variant is one number away

✓

Operation history intact — nothing collapsed or applied

LEAVE WITH A GRAPH YOU CAN CHANGE, NOT A MESH YOU CANNOT.

BEFORE NEXT CLASS

WATCH

U03 — Geometry from scratch. Point, line, edge, face, mesh, solid. The Blender lesson file for this one already exists; check the unit page.

BUILD

Keep going on P1a. It is due Week 4 — the sequence, the diagram and three variants.

WRITE

Your procedure as pseudocode, using the notation above. Read it aloud to someone who has not seen your model and fix what they cannot follow.

BRING

The editable file, and one question about something in it that behaves oddly.